KANGDA WEI

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RESEARCH INTEREST

Natural Language Processing, Large Language Model, LLM Reasoning, LLM Exploratory Thinking, Post-Training, Multi-Modal, Multi-Agent Systems

EDUCATION

Texas A&M University — College Station, TX
Doctor of Philosophy in Computer Science. Advisor: Ruihong Huang. — Aug 2023 - Dec 2027 (Expected)
University of North Carolina at Chapel-Hill — Chapel-Hill, NC
Master of Science in Computer Science. Advisor: Shashank Srivastava. — Aug 2022 - May 2023
University of North Carolina at Chapel-Hill — Chapel-Hill, NC
B.S. in Computer Science, B.S. in Statistics and Operational Research, GPA: 3.812/4.0 — Aug 2019 - May 2022

INDUSTRY EXPERIENCE

NLP & Large Language Model Intern
Robert Bosch LLC, USA — Sunnyvale, CA
Applied Scientist Intern, Mentor: Zhengyu Zhou — May 2024 - Aug 2024

RESEARCH EXPERIENCE

Enhancing Mathematical Reasoning in LLMs via Exploratory Thinking and Diversity-Aware GRPO
Texas A&M University — College Station, TX
Research Assistant, Mentor: Prof.Ruihong Huang — Present

- Developed a diversity-promoting reinforcement learning approach that integrates Maximal Marginal Relevance into GRPO to encourage exploration and reduce mode collapse during reasoning optimization.
- Adapted exploratory thinking principles to enhance mathematical reasoning performance across diverse problem-solving styles.
- Achieved an average of 1.5% absolute accuracy improvement across five mathematical reasoning benchmarks (AIME24, AMC23, MATH500, Minerva, and OlympiadBench) compared to both the original GRPO and other diversity-aware GRPO variants.
- *Ongoing:* Conducting ablation and generalization studies; paper under preparation.

Mitigating Bias and Improving Reasoning in LLMs via Exploratory Thinking and RL
Texas A&M University — College Station, TX
Research Assistant, Mentor: Prof.Ruihong Huang — Feb 2025 - Aug 2025

- Proposed a novel data generation framework that encourages exploratory thinking in LLMs by prompting them to reason over structurally identical, morally ambiguous story pairs with male and female protagonists.
- Designed a guided judgement reconciliation process to uncover and mitigate biased moral reasoning, producing balanced gender-neutral story-judgement pairs.
- Performed SFT and DPO on LLMs, achieving substantial bias reduction on WinoBias and GenMO benchmarks without sacrificing performance on MMLU and TruthfulQA.
- Published paper at *Findings of Empirical Methods in Natural Language Processing 2025.*
- *Ongoing:* Extending framework to other forms of social bias (e.g., race, social status).

Nuclear Safety Improvement via LLM
Texas A&M University — College Station, TX
Research Assistant, Mentor: Prof. Ruihong Huang — Oct 2024 – Present

- Collaborating with Idaho National Laboratory (INL) on a 3-year project to improve nuclear safety.
- Designed and implemented a multi-agent system that uses LLMs and VLMs to automatically generate nuclear fault trees via breadth-first search (BFS) expansion, enabling scalable analysis of system failures.
- Built a large-scale nuclear safety dataset by applying OCR to technical documents, generating structured questions with AI models, and validating results with domain experts.
- Evaluated and benchmarked multiple LLMs/VLMs for accuracy, and interpretability in nuclear safety tasks.
- Preparing research publications based on project outcomes.

LegalCore: A Dataset for Event Coreference Resolution in Legal Documents
Texas A&M University — College Station, TX
Research Assistant, Mentor: Prof.Ruihong Huang — Nov 2024 - Feb 2025

- Introduced LegalCore, the first dataset for the legal domain that has been annotated with comprehensive event and event coreference information.
- Benchmarked mainstream LLMs performance on LegalCore, finding that both event detection and event coreference remain challenging to LLMs.
- Provided a strong supervised baseline by fine-tuning small language models and conduct empirical analysis to inspire future works.
- Published paper at *Findings of the Empirical Methods in Natural Language Processing 2025*.

Multi-Agent Video Understanding for Advanced Indexing of Lecture Videos
Robert Bosch Research and Technology Center North America — Sunnyvale, CA
NLP & LLMs Intern, Mentor: Zhengyu Zhou — May 2024 - Sep 2024

- Proposed a novel end-to-end multi-agent multi-modal framework that leverages various large models for advanced understanding/indexing of presentation-style videos.
- The framework first segments each video into slide-presentation segments, leveraging vision-language model (VLM) to assist modern shot-detection approach.
- The framework then proceeds to understand each detected segment, generating multi-modal understanding results as indexes incorporating multiple agents, which capture rich and reliable multi-modal information to be used for downstream tasks in a salable way.
- Conducted systematic evaluations (intrinsic evaluation through Amazon Mechanical Turk, and extrinsic evaluation on open-ended QA) on the public LPM dataset as well as an internal enterprise video dataset, demonstrating the effectiveness of the proposed framework.
- Paper in submission. Patent submitted to the United States Patent and Trademark Office.

Are LLMs Good Annotators for Discourse-level Event Relation Extraction?
Texas A&M University — College Station, TX
Research Assistant, Mentor: Prof.Ruihong Huang — Aug 2023 - Apr 2024

- Assessed the effectiveness of different close-source and open-source LLMs, including GPT-3.5, GPT-4, and LLaMA-2, in addressing discourse-level ERE tasks characterized by lengthy documents and intricate relations encompassing coreference, temporal, causal, and subevent types.
- Revealed a notable underperformance of GPT models compared to the baseline established through supervised learning.
- Conducted quantitative and qualitative analysis to show that GPT model benefits from dividing intricate tasks into smaller components, however, still encounters challenges in avoiding hallucinations, satisfying transitivity rules among predictions, and capturing dependencies over long distances.
- Published paper at *Findings of the Empirical Methods in Natural Language Processing 2024*.

When Do Decompositions Help for Machine Reading?
Johns Hopkins University — Baltimore, MD
Visiting Research Assistant, Mentor: Prof.Benjamin van Durme — May 2022 - May 2023

- Explored the effect of decomposition on machine reading with an exhaustive set of variants across a range of models over the high-level Question Decomposition Meaning Representation (QDMR) BREAK dataset using various LLMs, including T-5, Alpaca, and LLaMA.
- Discovered that question decomposition is not helpful for machine reading but rather harmful.
- Conducted a qualitative error analysis, showing that machine reading using question decomposition struggle due to compound error and question decomposition in bad formats.
- Published paper at *Proceedings of the Empirical Methods in Natural Language Processing 2023*.

Leveraging Multiple Teachers for Test-Time Adaptation of Language-Guided Classifiers
University of North Carolina at Chapel Hill — Chapel Hill, NC
Research Assistant, Mentor: Prof.Shashank Srivastava — Apr 2022 - May 2023

- Present a framework for test-time adaptation of language explanation-guided classifiers towards a specific task during inference.
- Achieved 8% higher classification accuracy by utilizing label aggregation with language models, including RoBERTa, T0, OPT, and Flan-T5, for test-time adaptation, and three times better accuracy comparing to baselines with zero-shot learning.
- Further improved the accuracy by 20% by incorporating self-learning by fine-tuning pre-trained LM on noisy labeled data.
- Conducted qualitative analysis for framework's interpretability.
- Simplify and improve the label aggregation technique by replacing hand-written labeling functions with LM.

- Published paper at *Findings of the Empirical Methods in Natural Language Processing 2023.*

Compositional Generalization for Kinship Prediction through Data Augmentation
University of North Carolina at Chapel Hill Chapel Hill, NC
Research Assistant, Mentor: Prof.Shashank Srivastava Feb 2021 - Mar 2022

- Evaluated empirically the utility of data augmentation and intermediate structured representations towards compositional generalization for the task of kinship prediction from a story.
- Tested the impact of incorporating data augmentation and intermediate structured data on model's performance. Data augmentation boosted generalization performance by around 20% on average relative to a baseline model from prior work.
- Found that predicting and using intermediate kinship graphs led to a deterioration in the generalization of kinship prediction.
- Published paper at *Proceedings of the 4th Workshop of Narrative Understanding (WNU2022).*

A Multilingual COVID-19 Question Answering System
University of California, Santa Barbara Santa Barbara, CA
Visiting Research Assistant, Mentor: Prof. William Wang May 2021 - Aug 2021

- Established a multilingual COVID-19 Question Answering system using mBERT and XLM-Roberta models.
- Focused primarily on building and training the Reading Comprehension part of the QA system, including collecting
- Performed large-scale neural machine translation on entire CORD-19 dataset to acquire data in Chinese and French to alleviate the data scarcity problem in foreign languages.
- Reached an F1 score of 60.5 for the reading comprehension model of the final QA system.

Temporal and Spatial Analysis on U.S. Domestic Mobility
Research Assistant Chapel Hill, NC
Mentor: Prof. Weizi Li at the University of Memphis, Prof.Lei Lin at University of Rochester Jun 2020 - Oct 2020

- Investigated the impact of COVID-19 on the United States mobility through temporal and spatial analysis with network models.
- Collected relative datasets from GitHub, Kaggle, New York Times and Google Search Trends; Preprocessed data with Python and implemented the model with Networkx package.
- Published paper in the 20th and 21st Joint COTA International Conference of Transportation Professionals.

PUBLICATIONS

- Cheng Zhang, Rajasekhar Kakarla, **Kangda Wei**, Ruihong Huang. ENG-DRB: PDTB-style Discourse Relation Bank on Engineering Tutorial Video Scripts. *In submission.*

- **Kangda Wei**, Hasnat Md Abdullah, Ruihong Huang. Mitigating Gender Bias via Fostering Exploratory Thinking in LLMs. *Findings of the Empirical Methods in Natural Language Processing 2025.*

- Abhilekh Borah, Hasnat Md Abdullah, **Kangda Wei**, Ruihong Huang. CliME: Evaluating Multimodal Climate Discourse on Social Media and the Climate Alignment Quotient (CAQ). *NLP4PosImpact Workshop, Association for Computational Linguistics 2025. Best Paper Award.*

- **Kangda Wei**, Xi Shi, Jonathan Tong, Sai Ramana Reddy, Anandhavelu Natarajan, Rajiv Jain, Aparna Garimella, Ruihong Huang. LegalCore: A Dataset for Event Coreference Resolution in Legal Documents. *Findings of Association for Computational Linguistics 2025.*

- **Kangda Wei**, Zhengyu Zhou, Bingqing Wang, Jun Araki, Lukas Lange, Ruihong Huang, Zhe Feng. Multi-Agent Video Understanding for Advanced Indexing of Lecture Videos. *In submission.*

- Hasnat Md Abdullah, Tian Liu, **Kangda Wei**, Shu Kong, Ruihong Huang. UALBench: The First Comprehensive Unusual Activity Localization Benchmark. *Winter Conference on Applications of Computer Vision (WACV) 2025.*

- **Kangda Wei**, Aayush Guatam, Ruihong Huang. Are LLMs Good Annotators for Discourse-level Event Relation Extraction? *Findings of the Empirical Methods in Natural Language Processing 2024.*

- Jiangshu Du, Yibo Wang, Wenting Zhao, ... **Kangda Wei**, et al. LLMs Assist NLP Researchers: Critique Paper (Meta-)Reviewing. *Proceedings of the Empirical Methods in Natural Language Processing 2024.*

- **Kangda Wei**, Dawn Lawrie, Benjamin Van Durme, Yunmo Chen, Orion Weller. When Do Decompositions Help for Machine Reading? *Proceedings of the Empirical Methods in Natural Language Processing 2023*

- **Kangda Wei**, Sayan Ghosh, Rakesh Menon, and Shashank Srivastava. Leveraging Multiple Teachers for Test-Time Adaptation of Language-Guided Classifiers. *Findings of the Empirical Methods in Natural Language Processing 2023*

- **Kangda Wei**, Sayan Ghosh, and Shashank Srivastava. 2022. Compositional Generalization for Kinship Prediction through Data Augmentation. *Proceedings of the 4th Workshop of Narrative Understanding (WNU2022)*, pages 1319, Seattle, United States. Association for Computational Linguistics.

- *Songhe Wang, ***Kangda Wei**, Lei Lin, Weizi Li. Spatial-temporal Analysis of COVID-19's Impact on Human Mobility: the Case of the United States, in the *20th and 21st Joint COTA International Conference of Transportation Professionals.* *Co-author: equal contribution

PROFESSIONAL SERVICES

- Reviewer for ACL Rolling Review (ARR) since February 2024 (covering ACL/NAACL/EMNLP/EACL/AACL).
- Reviewer for IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2025.
- Reviewer for The Computer Journey.

TEACHING SERVICES

Graduate Teaching Assistant:

- COMP211 System Fundamentals, UNC-Chapel Hill, Department of Computer Science, Spring 2023
- COMP431 Internet Services and Protocols, UNC-Chapel Hill, Department of Computer Science, Fall 2022

AWARDS

- Texas A&M University CSE Department Travel Grant, 2023, 2024

SKILLS

- **Programming Languages / Frameworks:** Python, Java, JavaScript, C, HTML, CSS, MATLAB, R, PyTorch, JAX, Hugging Face Transformers, Accelerate, Linux, Parallel Model Training
- **Software & Tools:** PyCharm, VS Code, Spyder, Jupyter Notebook, Weights & Biases,, Docker, Slurm, Git, Tableau, LaTeX, Anaconda, RStudio
- **LLM Training & Post-training:** SFT, RLHF, DPO, GRPO, Reward Modeling, Prompt Engineering, Synthetic Data Generation
- **Cloud & Infrastructure:** AWS, Microsoft Azure, Amazon Mechanical Turk
- **Language:** Chinese (native), English (proficient)